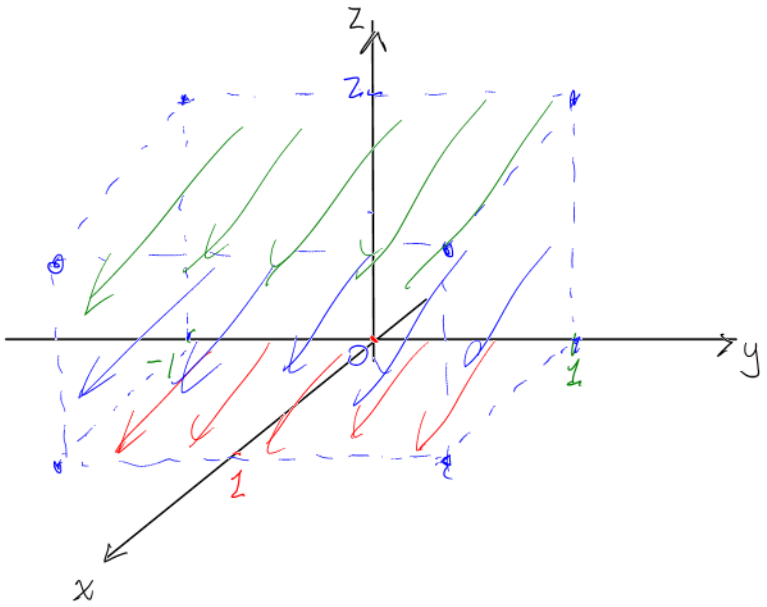
MA1521 L1 Week 10 Mon 20261019

Evaluate: $\int_0^2 \int_{-1}^1 \int_0^1 (x+y+z)\,dx\,dy\,dz$



$$\int_0^2 \int_{-1}^1 \int_0^1 (x+y+z)\,dx\,dy\,dz$$

$$= \int_0^2 \int_{-1}^1 \left[\frac{x^2}{2} + xy + xz\right]_{x=0}^{x=1} dy\,dz$$

$$= \int_0^2 \int_{-1}^1 \left(\frac{1}{2} + y + z\right) dy\,dz$$

$$= \int_0^2 \left[\frac{y}{2} + \frac{y^2}{2} + yz\right]_{y=-1}^{y=1} dz$$

$$= \int_0^2 (1+2z)\,dz = \left[z + z^2\right]_{z=0}^{z=2} = 2 + 2^2 = 6$$

Evaluate: $\displaystyle\int_0^2 \int_0^{\sqrt{y}} \int_0^{xy} 2xy^2 z \, dz \, dx \, dy$

$$= \int_0^2 \int_0^{\sqrt{y}} 2xy^2 \left[\frac{z^2}{2}\right]_{z=0}^{z=xy} dx\, dy$$

$$= \int_0^2 \int_0^{\sqrt{y}} xy^2 \left(x^2y^2 - 0\right) dx\, dy = \int_0^2 \int_0^{\sqrt{y}} x^3 y^4 \, dx\, dy$$

$$= \int_0^2 y^4 \left[\frac{x^4}{4}\right]_{x=0}^{x=\sqrt{y}} dy$$

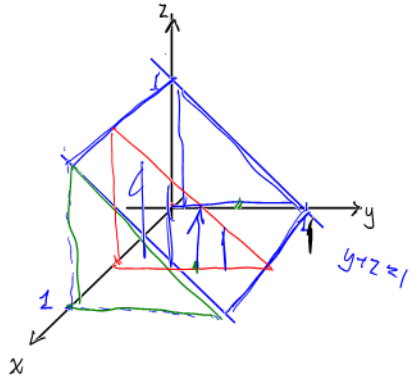
$$= \int_0^2 \frac{y^6}{4} dy = \frac{1}{4}\left[\frac{y^7}{7}\right]_{y=0}^{y=2}$$

$$= \frac{2^7}{4 \cdot 7} = \frac{32}{7}$$

y- & z- axes

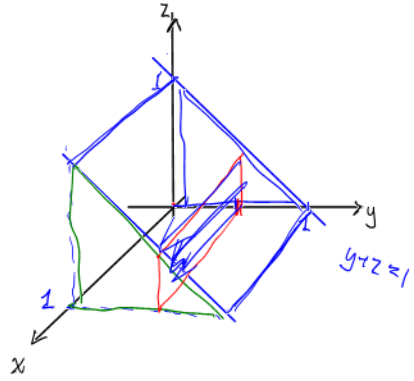
The rectangular prism having as its two bases the triangle in the yz-plane cut out by the two axes and the line $y + z = 1$, and the corresponding triangle in the plane $x = 1$ obtained by adding 1 to the x-coordinate of each point in the first triangle. Supply limits for three different orders of integration:

(i) $\iiint dz\,dy\,dx$



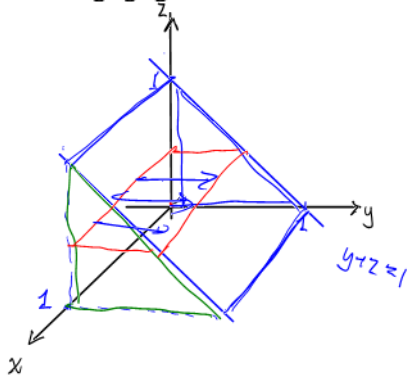
$$\int_{x=0}^{x=1}\int_{y=0}^{y=1}\int_{z=0}^{z=1-y} dz\,dy\,dx$$

(ii) $\iiint dx\,dz\,dy$



$$\int_{y=0}^{y=1}\int_{z=0}^{z=1-y}\int_{x=0}^{x=1} dx\,dz\,dy$$

(iii) $\iiint dy\,dx\,dz$

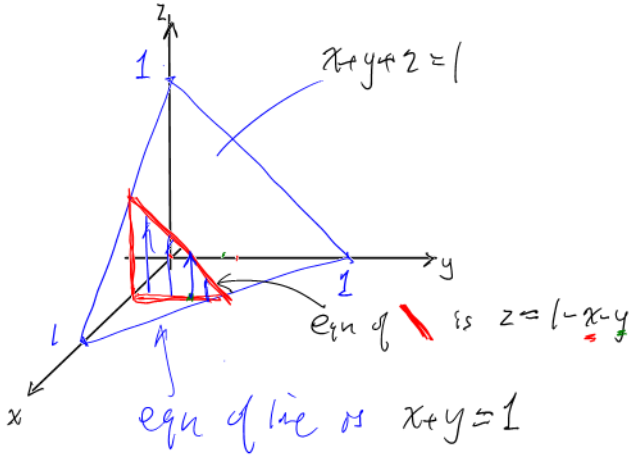


$$\int_{z=0}^{z=1}\int_{x=0}^{x=1}\int_{y=0}^{y=1-z} dy\,dx\,dz$$

$x \geq 0,\ y \geq 0,\ z \geq 0$

yz-plane $x = 0$,
xz-plane $y = 0$,
xy-plane $z = 0$,

Find the center of mass of the tetrahedron D in the first octant formed by the coordinate planes and the plane $x + y + z = 1$. Assume the density $\delta = 1$.



Compute $\left(\dfrac{\iiint_D x\,dV}{\iiint_D 1\,dV}, \dfrac{\iiint_D y\,dV}{\iiint_D 1\,dV}, \dfrac{\iiint_D z\,dV}{\iiint_D 1\,dV} \right)$

$\underbrace{\qquad}_{x_0} \quad \underbrace{\qquad}_{y_0} \quad \underbrace{\qquad}_{z_0}$

Volume of D $= \iiint_D 1\,dV = \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} \int_{z=0}^{z=1-x-y} 1\,dz\,dy\,dx$

$= \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} (1-x-y)\,dy\,dx$

$= \int_{x=0}^{x=1} \left[y - xy - \frac{y^2}{2} \right]_{y=0}^{y=1-x} dx \qquad = \int_{x=0}^{x=1} \frac{1}{2}(1-2x+x^2)\,dx$

$= \frac{1}{2}\left[x - x^2 + \frac{x^3}{3} \right]_{x=0}^{x=1} = \frac{1}{2}\left(1 - 1 + \frac{1}{3}\right) = \frac{1}{6}$

By symmetry in x, y, z, must have $\iiint_D x\,dV = \iiint_D y\,dV = \iiint_D z\,dV$

$$\iiint_D x\,dV = \int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y} x\,dz\,dy\,dx$$

$$= \int_{x=0}^{x=1} x \underbrace{\left(\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y} 1\,dz\,dy\right)}_{h} dx$$

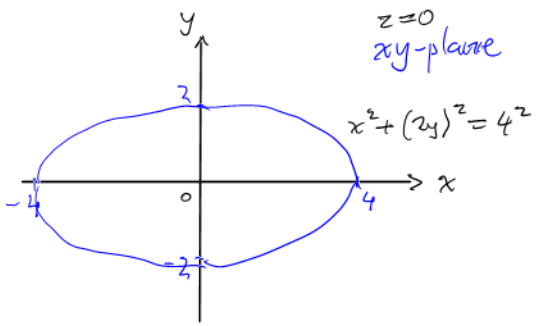
h $\frac{1}{2}(1-2x+x^2)$ from above!

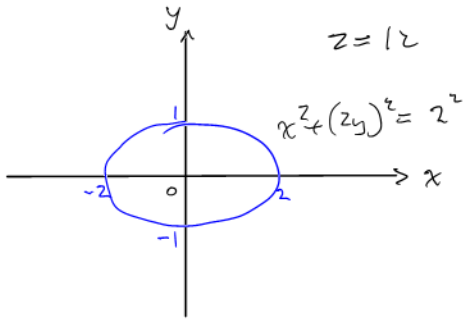
$$= \int_{x=0}^{x=1} \frac{1}{2}(x-2x^2+x^3)\,dx = \frac{1}{2}\left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right]_{x=0}^{x=1}$$

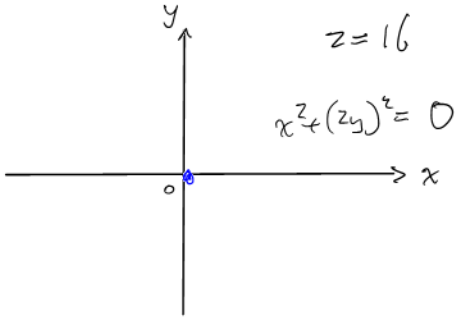
$$= \frac{1}{2}\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \boxed{\frac{1}{24}}$$

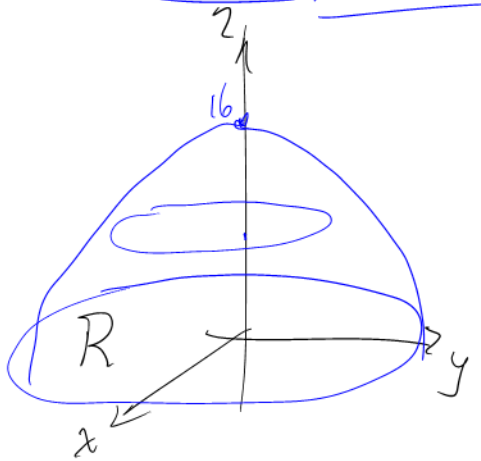
So center of mass of D is $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$

Find the volume underneath the surface $z = 16 - x^2 - 4y^2$ and over the xy-plane:









Volume $:= \iiint_R 1 \, dV$

$= \int_{z=0}^{z=16} \iint_{R_z} 1 \, dx\, dy \, dz$

$R_z :=$ ellipse $x^2+(2y)^2 = 16-z$

Fix z. $\in [0,16]$

$\iint_{R_z} 1 \, dx\, dy$

$R_z :=$ ellipse $x^2+(2y)^2 \le 16-z$ (≥ 0)

$=$ Ch. of Var.

$u = x$

$v = 2y$

$J(x,y) = \det\begin{pmatrix} 1 & \\ & 2 \end{pmatrix} = 2$

$\iint_{D_z} \frac{1}{2} \, du\, dv$

$D_z =$ circular disc $u^2+v^2 \le 16-z$

$=$ polar coords in u,v

$\frac{1}{2} \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=\sqrt{16-z}} r \, dr\, d\theta$

$$= \pi \cdot \left[\frac{r^2}{2} \right]_{r=0}^{r=\sqrt{16-z}}$$

$$= \frac{\pi}{2} \cdot (16 - z)$$

$$\text{Volume} = \int_{z=0}^{z=16} \frac{\pi}{2} \cdot (16 - z)\, dz = \frac{\pi}{2} \left[16z - \frac{z^2}{2} \right]_{z=0}^{z=16}$$

$$= \frac{\pi}{2} \left(16^2 - \frac{16^2}{2} \right) = 64\pi$$